Computations for the Automated Pigpen Washing System

1. Given Data

Canonical givens from the Computation for the Improved Design. Gauge pressures; floor is the datum.

Parameter	Symbol	Value
Water density	ρ	$1000\ \text{kg/m}^3$
Dynamic viscosity	μ	$0.001002\ \text{kg/m·s}$
Gravity	g	$9.81\ \text{m/s}^2$
Specific weight	γ	$9{,}810\ \text{N/m}^3$
Absolute roughness (PPR and hose)	ε	$1.5\times10^{-6}\ \text{m}$
Pump rated flow	Q	$17.5\ \text{L/min} = 2.917\times10^{-4}\ \text{m}^3/\text{s}$
Pump pressure (relief setting)	P_1	$35\ \text{kgf/cm}^2 = 3{,}432{,}327.5\ \text{Pa}$
1¼″ pipe inside diameter	$D_{1¼}$	$0.0345\ \text{m}$
¾″ pipe inside diameter	$D_{¾p}$	$0.0215\ \text{m}$
¾″ hose inside diameter	$D_{¾h}$	$0.01905\ \text{m}$
Nozzle orifice diameter / discharge coefficient	d_o, C_d	$0.001\ \text{m}$ / 0.60 (assumed)
Nozzles per branch	N	Left 6, Right 6, Front 6, Top 1
Pump outlet elevation	z_1	$0.322\ \text{m}$
Nozzle elevation	z_2	Walls $1.070\ \text{m}$; Top $2.059\ \text{m}$
Pump efficiency	η	65%
Operating time	t	$30\ \text{min} = 1800\ \text{s}$

2. Cross-Sectional Area

Formula:

$$A = \pi D^2/4$$

$$A_{1¼″\ \text{pipe}} = \pi(0.0345\ \text{m})^2/4 = 9.34820\times10^{-4}\ \text{m}^2$$

$$A_{¾″\ \text{pipe}} = \pi(0.0215\ \text{m})^2/4 = 3.63050\times10^{-4}\ \text{m}^2$$

$A_{¾″\ hose} = \pi(0.01905\ \text{m})^2/4 = 2.85023\times10^{-4}\ \text{m}^2$

3. Flow Rate

Formula, with the branch flow divided equally among its nozzles and reduced by one nozzle flow after each tee:

$Q_{nozzle} = Q/N$

Top branch (1 nozzle): $Q = 2.917\times10^{-4}\ \text{m}^3/\text{s}$

Left, Right, Front (6 nozzles): $Q_{nozzle} = 2.917\times10^{-4}/6 = 4.86167\times10^{-5}\ \text{m}^3/\text{s}$

Flow in the ¾″ nozzle pipe segments:

Nozzles 3 and 4: $Q_3 = Q/2 = 2.917\times10^{-4}/2 = 1.45850\times10^{-4}\ \text{m}^3/\text{s}$

Nozzles 2 and 5: $Q_2 = Q_3 - Q_{nozzle} = 1.45850\times10^{-4} - 4.86167\times10^{-5} = 9.72333\times10^{-5}\ \text{m}^3/\text{s}$

Nozzles 1 and 6: $Q_1 = Q_2 - Q_{nozzle} = 9.72333\times10^{-5} - 4.86167\times10^{-5} = 4.86167\times10^{-5}\ \text{m}^3/\text{s}$

4. Velocity

Formula:

$v = Q/A$

1¼″ pipe: $v = 2.917\times10^{-4}/9.34820\times10^{-4} = 0.312039\ \text{m/s}$

¾″ pipe: $v = 2.917\times10^{-4}/3.63050\times10^{-4} = 0.803470\ \text{m/s}$

¾″ hose: $v = 2.917\times10^{-4}/2.85023\times10^{-4} = 1.02343\ \text{m/s}$

Nozzles 3 and 4: $v = 1.45850\times10^{-4}/3.63050\times10^{-4} = 0.401735\ \text{m/s}$

Nozzles 2 and 5: $v = 9.72333\times10^{-5}/3.63050\times10^{-4} = 0.267823$ m/s

Nozzles 1 and 6: $v = 4.86167\times10^{-5}/3.63050\times10^{-4} = 0.133912$ m/s

5. Reynolds Number

Formula:

$$Re = \rho v D/\mu$$

1¼″ pipe: $Re = (1000)(0.312039)(0.0345)/0.001002 = 10{,}743.84$

¾″ hose: $Re = (1000)(1.02343)(0.01905)/0.001002 = 19{,}457.36$

¾″ pipe, full flow: $Re = (1000)(0.803470)(0.0215)/0.001002 = 17{,}240.12$

¾″ pipe, nozzle 3: $Re = (1000)(0.401735)(0.0215)/0.001002 = 8620.062$

¾″ pipe, nozzle 2: $Re = (1000)(0.267823)(0.0215)/0.001002 = 5746.708$

¾″ pipe, nozzle 1: $Re = (1000)(0.133912)(0.0215)/0.001002 = 2873.354$

6. Friction Factor (Haaland)

Formula:

$$f = [-1.8\log_{10}((\varepsilon/D/3.7)^{1.11} + 6.9/Re)]^{-2}$$

1¼″ pipe: $\varepsilon/D = (1.5\times10^{-6})/(0.0345) = 4.34783\times10^{-5}$; $\varepsilon/D/3.7 = 1.17509\times10^{-5}$; $(\varepsilon/D/3.7)^{1.11} = 3.37115\times10^{-6}$

$6.9/Re = 6.9/10{,}743.84 = 6.42228\times10^{-4}$; $sum = 6.45599\times10^{-4}$; $\log_{10}(sum) = -3.19004$

$f = [-1.8(-3.19004)]^{-2} = 0.0303294$

¾″ hose: $\varepsilon/D = (1.5\times10^{-6})/(0.01905) = 7.87402\times10^{-5}$; $\varepsilon/D/3.7 = 2.12811\times10^{-5}$; $(\varepsilon/D/3.7)^{1.11} = 6.51739\times10^{-6}$

$6.9/Re = 6.9/19{,}457.36 = 3.54622\times10^{-4}$; sum $= 3.61139\times10^{-4}$; $\log_{10}(\text{sum}) = -3.44233$

$f = [-1.8(-3.44233)]^{-2} = 0.0260466$

¾″ pipe, full flow: $\varepsilon/D = (1.5\times10^{-6})/(0.0215) = 6.97674\times10^{-5}$; $\varepsilon/D/3.7 = 1.88561\times10^{-5}$; $(\varepsilon/D/3.7)^{1.11} = 5.69837\times10^{-6}$

$6.9/Re = 6.9/17{,}240.12 = 4.00229\times10^{-4}$; sum $= 4.05928\times10^{-4}$; $\log_{10}(\text{sum}) = -3.39155$

$f = [-1.8(-3.39155)]^{-2} = 0.0268323$

¾″ pipe, nozzle 3: $\varepsilon/D = (1.5\times10^{-6})/(0.0215) = 6.97674\times10^{-5}$; $\varepsilon/D/3.7 = 1.88561\times10^{-5}$; $(\varepsilon/D/3.7)^{1.11} = 5.69837\times10^{-6}$

$6.9/Re = 6.9/8620.062 = 8.00458\times10^{-4}$; sum $= 8.06157\times10^{-4}$; $\log_{10}(\text{sum}) = -3.09358$

$f = [-1.8(-3.09358)]^{-2} = 0.0322502$

¾″ pipe, nozzle 2: $\varepsilon/D = (1.5\times10^{-6})/(0.0215) = 6.97674\times10^{-5}$; $\varepsilon/D/3.7 = 1.88561\times10^{-5}$; $(\varepsilon/D/3.7)^{1.11} = 5.69837\times10^{-6}$

$6.9/Re = 6.9/5746.708 = 0.00120069$; sum $= 0.00120639$; $\log_{10}(\text{sum}) = -2.91851$

$f = [-1.8(-2.91851)]^{-2} = 0.0362353$

¾″ pipe, nozzle 1: $\varepsilon/D = (1.5\times10^{-6})/(0.0215) = 6.97674\times10^{-5}$; $\varepsilon/D/3.7 = 1.88561\times10^{-5}$; $(\varepsilon/D/3.7)^{1.11} = 5.69837\times10^{-6}$

$6.9/Re = 6.9/2873.354 = 0.00240137$; sum = 0.00240707; $\log_{10}(\text{sum}) = -2.61851$

$f = [-1.8(-2.61851)]^{-2} = 0.0450139$

7. Major Head Loss (Darcy–Weisbach)

Formula:

$h_f = f(L/D)(v^2/2g)$

Pipe lengths per branch (1¼″ pipe):

Front: $L = 1.244 + 1.870 + 1.313 + 0.074 + 1.273 + 0.154 = 5.928$ m

Left and Right: $L = 1.244 + 1.870 + 1.313 + 0.074 + 0.915 = 5.416$ m

Top: $L = 1.244 + 1.870 + 1.313 + 0.074 + 0.249 = 4.750$ m

Hose: 2.0 m (Front, Left, Right), 1.5 m (Top). ¾″ pipe: 0.061 m (full flow), 0.197 m (nozzle 3), 0.395 m (nozzle 2), 0.380 m (nozzle 1).

Front branch:

$h_{f,1¼} = (0.030329)(5.928/0.0345)(0.312039)^2/(2\cdot9.81) = 0.0258625$ m

$h_{f,hose} = (0.026047)(2/0.01905)(1.02343)^2/(2\cdot9.81) = 0.145982$ m

$h_{f,¾p} = (0.026832)(0.061/0.0215)(0.803470)^2/(2\cdot9.81) = 0.00250490$ m

$h_{f,n3} = (0.032250)(0.197/0.0215)(0.401735)^2/(2\cdot9.81) = 0.00243075$ m

$h_{f,n2} = (0.036235)(0.395/0.0215)(0.267823)^2/(2\cdot9.81) = 0.00243382$ m

$$h_{f,n1} = (0.045014)(0.38/0.0215)(0.133912)^2/(2\cdot 9.81) = 7.27159\times10^{-4}\ \text{m}$$

$$h_f = 0.025862 + 0.14598 + 0.0025049 + 0.0024307 + 0.0024338 + 7.2716\times10^{-4} = 0.179941\ \text{m}$$

Left and Right branches (identical):

$$h_{f,1¼} = (0.030329)(5.416/0.0345)(0.312039)^2/(2\cdot 9.81) = 0.0236288\ \text{m}$$

$$h_{f,hose} = (0.026047)(2/0.01905)(1.02343)^2/(2\cdot 9.81) = 0.145982\ \text{m}$$

$$h_{f,¾p} = (0.026832)(0.061/0.0215)(0.803470)^2/(2\cdot 9.81) = 0.00250490\ \text{m}$$

$$h_{f,n3} = (0.032250)(0.197/0.0215)(0.401735)^2/(2\cdot 9.81) = 0.00243075\ \text{m}$$

$$h_{f,n2} = (0.036235)(0.395/0.0215)(0.267823)^2/(2\cdot 9.81) = 0.00243382\ \text{m}$$

$$h_{f,n1} = (0.045014)(0.38/0.0215)(0.133912)^2/(2\cdot 9.81) = 7.27159\times10^{-4}\ \text{m}$$

$$h_f = 0.023629 + 0.14598 + 0.0025049 + 0.0024307 + 0.0024338 + 7.2716\times10^{-4} = 0.177708\ \text{m}$$

Top branch:

$$h_{f,1¼} = (0.030329)(4.75/0.0345)(0.312039)^2/(2\cdot 9.81) = 0.0207232\ \text{m}$$

$$h_{f,hose} = (0.026047)(1.5/0.01905)(1.02343)^2/(2\cdot 9.81) = 0.109487\ \text{m}$$

$$h_{f,Top} = 0.020723 + 0.10949 = 0.130210\ \text{m}$$

8. Minor Head Loss

Formula:

$$h_m = K(v^2/2g)$$

Left and Right branches. ΣK: 1¼″ = 1 + 0.10 + 2 + 0.10 = 3.2; hose = 0.3 + 10 + 0.10 = 10.4; ¾″ pipe = 1.90 + 0.10 = 2; nozzles 3, 2, 1 = 0.5, 0.5, 2.

$$h_{m,1¼} = (3.2)(0.312039)^2/(2 \cdot 9.81) = 0.015881 \text{ m}$$

$$h_{m,hose} = (10.4)(1.02343)^2/(2 \cdot 9.81) = 0.55520 \text{ m}$$

$$h_{m,¾p} = (2)(0.803470)^2/(2 \cdot 9.81) = 0.065807 \text{ m}$$

$$h_{m,n3} = (0.5)(0.401735)^2/(2 \cdot 9.81) = 0.0041129 \text{ m}$$

$$h_{m,n2} = (0.5)(0.267823)^2/(2 \cdot 9.81) = 0.0018280 \text{ m}$$

$$h_{m,n1} = (2)(0.133912)^2/(2 \cdot 9.81) = 0.0018280 \text{ m}$$

$$h_m = 0.015881 + 0.55520 + 0.065807 + 0.0041129 + 0.0018280 + 0.0018280 = 0.644654 \text{ m}$$

Front branch. ΣK: 1¼″ = 1 + 0.10 + 0.5 + 0.10 + 0.5 = 2.2; hose, ¾″ pipe and nozzles as above.

$$h_{m,1¼} = (2.2)(0.312039)^2/(2 \cdot 9.81) = 0.010918 \text{ m}$$

$$h_{m,hose} = (10.4)(1.02343)^2/(2 \cdot 9.81) = 0.55520 \text{ m}$$

$$h_{m,¾p} = (2)(0.803470)^2/(2 \cdot 9.81) = 0.065807 \text{ m}$$

$$h_{m,n3} = (0.5)(0.401735)^2/(2 \cdot 9.81) = 0.0041129 \text{ m}$$

$$h_{m,n2} = (0.5)(0.267823)^2/(2 \cdot 9.81) = 0.0018280 \text{ m}$$

$$h_{m,n1} = (2)(0.133912)^2/(2 \cdot 9.81) = 0.0018280 \text{ m}$$

$$h_m = 0.010918 + 0.55520 + 0.065807 + 0.0041129 + 0.0018280 + 0.0018280 = 0.639691 \text{ m}$$

Top branch. ΣK: 1¼″ = 1 + 0.10 + 0.5 + 1.8 = 3.4; hose = 0.3 + 10 + 0.10 + 0.10 = 10.5.

$h_{m,1¼} = (3.4)(0.312039)^2/(2 \cdot 9.81) = 0.016873$ m

$h_{m,hose} = (10.5)(1.02343)^2/(2 \cdot 9.81) = 0.56054$ m

$h_{m,Top} = 0.016873 + 0.56054 = 0.577409$ m

9. Total Head Loss

$h_{total} = h_f + h_m$

Left and Right: $h_{total} = 0.177708 + 0.644654 = 0.822361$ m

Front: $h_{total} = 0.179941 + 0.639691 = 0.819632$ m

Top: $h_{total} = 0.130210 + 0.577409 = 0.707619$ m

10. Nozzle Inlet Pressure (Energy Equation)

Formula, from the pump outlet (1) to the nozzle inlet (2):

$P_2 = P_1 - \gamma[(v_{hose}{}^2 - v_{1¼}{}^2)/2g + (z_2 - z_1) + h_{total}]$

Velocity head term: $(1.02343^2 - 0.312039^2)/(2 \cdot 9.81) = 0.048422$ m

Left and Right: $P_2 = 3{,}432{,}327.5 - 9{,}810[0.048422 + (1.07 - 0.322) + 0.822361]$

$P_2 = 3{,}432{,}327.5 - 9{,}810(1.61878) = 3{,}416{,}447.24$ Pa

Front: $P_2 = 3{,}432{,}327.5 - 9{,}810[0.048422 + (1.07 - 0.322) + 0.819632]$

$P_2 = 3{,}432{,}327.5 - 9{,}810(1.61605) = 3{,}416{,}474.01$ Pa

Top: $P_2 = 3{,}432{,}327.5 - 9{,}810[0.048422 + (2.059 - 0.322) + 0.707619]$

$P_2 = 3{,}432{,}327.5 - 9{,}810(2.49304) = 3{,}407{,}870.77$ Pa

11. Nozzle Discharge and Spray

Orifice area and discharge from the nozzle inlet pressure, with $C_d = 0.60$ (analysis):

$A_o = \pi(0.001)^2/4 = 7.85398\times10^{-7}$ m^2

$v_{jet} = \sqrt{(2P_2/\rho)}$; $Q_o = C_d A_o v_{jet}$; $F = \rho Q_o v_{jet}$

Left and Right: $v_{jet} = \sqrt{(2\cdot3{,}416{,}447.24/1000)} = 82.661$ m/s

$Q_o = (0.60)(7.8540\times10^{-7})(82.661) = 3.8953\times10^{-5}$ m^3/s $= 2.3372$ L/min

Branch flow $= 6(2.3372) = 14.023$ L/min; relief return $= 17.5 - 14.023 = 3.477$ L/min

$F = (1000)(3.8953\times10^{-5})(82.661) = 3.220$ N per nozzle; branch $= 6(3.220) = 19.32$ N

Front: $v_{jet} = \sqrt{(2\cdot3{,}416{,}474.01/1000)} = 82.662$ m/s

$Q_o = (0.60)(7.8540\times10^{-7})(82.662) = 3.8953\times10^{-5}$ m^3/s $= 2.3372$ L/min

Branch flow $= 6(2.3372) = 14.023$ L/min; relief return $= 17.5 - 14.023 = 3.477$ L/min

$F = (1000)(3.8953\times10^{-5})(82.662) = 3.220$ N per nozzle; branch $= 6(3.220) = 19.32$ N

Top: $v_{jet} = \sqrt{(2\cdot3{,}407{,}870.77/1000)} = 82.558$ m/s

$Q_o = (0.60)(7.8540\times10^{-7})(82.558) = 3.8904\times10^{-5}$ m^3/s $= 2.3343$ L/min

Branch flow $= 1(2.3343) = 2.3343$ L/min; relief return $= 17.5 - 2.3343 =$ 15.17 L/min

$F = (1000)(3.8904\times10^{-5})(82.558) = 3.212$ N per nozzle; branch $= 1(3.212) =$ 3.212 N

Check of the canonical flow (2.917×10^{-4} m^3/s per branch) against the orifice. Required nozzle pressure for one wall nozzle at Q_{nozzle}:

$P_{req} = (\rho/2)[Q_{nozzle}/(C_d A_o)]^2 = 500[4.86167\times10^{-5}/((0.60)(7.8540\times10^{-7}))]^2 =$ $5{,}321{,}794$ Pa $= 5.32$ MPa

$P_{req}/P_1 = 1.550$

Spray angle for neighboring sprays to touch the target wall (full angle):

$\theta = 2\tan^{-1}(p/2s)$

Left and Right: $\theta = 2\tan^{-1}(0.39468/(2\cdot2.0)) = 11.27°$

Front: $\theta = 2\tan^{-1}(0.20527/(2\cdot2.5)) = 4.702°$

12. Hydraulic Power and Pump Input Power

Formulas:

$H = P_1/\gamma + z_1 + v_{1¼}^2/2g$; $P_h = \rho g Q H$; $P_{pump} = P_h/\eta$

$H = 3{,}432{,}327.5/9{,}810 + 0.322 + (0.312039)^2/(2\cdot9.81) = 349.880 + 0.322 +$ $0.0049627 = 350.207$ m

$P_h = (1000)(9.81)(2.917\times10^{-4})(350.207) = 1002.15$ W

$P_{pump} = 1002.15/0.65 = 1541.76$ W $= 2.0675$ hp

Motor check against the 1.1 kW (1.5 hp) nameplate, and pressure-rating check against the PN20 PPR pipe (2.0 MPa):

$$P_h/1100 = 1002.15/1100 = 91.10\%; \quad P_{pump}/1100 = 1541.76/1100 = 140.2\%$$

$$P_1/P_{rated} = 3.4323/2.0 = 1.72$$

13. Water and Electrical Energy (30 min)

$$V = Q \cdot t = (2.917 \times 10^{-4})(1800) = 0.52506\ m^3$$

$$E = P_{pump} \cdot t = (1541.76\ W)(0.5\ h) = 770.881\ Wh$$

14. Summary of Results

Quantity	Left / Right	Front	Top
Major head loss, h_f (m)	0.177708	0.179941	0.130210
Minor head loss, h_m (m)	0.644654	0.639691	0.577409
Total head loss (m)	0.822361	0.819632	0.707619
Nozzle inlet pressure P_2 (Pa)	3,416,447.24	3,416,474.01	3,407,870.77
Jet velocity (m/s)	82.66	82.66	82.56
Branch flow at P_2 (L/min)	14.02	14.02	2.334

Quantity	Result
Hydraulic power, P_h	1002.15 W
Pump input power, P_{pump}	1541.76 W (2.0675 hp)
Water used in 30 min	0.52506 m^3
Electrical energy in 30 min	770.881 Wh